

INDIAN INSTITUTE OF TECHNOLOGY BOMBAY

Department of Energy Science and Engineering

Self-Initiated Project — Declaration Form

Student	Aarya Parekh
Roll number	24B1535
Department	Energy Science and Engineering (DESE)
Programme	B.Tech Energy Engineering
Semester	Autumn 2026-27
Project type	Self-initiated / Independent

Topic

Energy Stock Price Forecasting Using LSTM Neural Networks — a quantitative project applying deep learning to forecast the next-day closing price of NTPC. The project builds a complete ML pipeline: data collection via Yahoo Finance, exploratory data analysis, feature engineering (17 technical indicators), baseline modelling (Linear Regression, Random Forest), and LSTM training with early stopping. Implemented in Python (PyTorch, scikit-learn, pandas) and version-controlled on GitHub.

Scope and deliverables

#	Deliverable	Format
1	Working Python codebase (7 scripts)	.py files on GitHub
2	Trained LSTM model	.pth (PyTorch checkpoint)
3	9 analysis plots	.png files
4	Model comparison results	.json + README table
5	Project report	.pdf
6	README with setup instructions	.md on GitHub

Duration: 3 weeks (1 August – 21 August 2026)

GitHub: github.com/aariiparekh3012-collab/project1-aarya

I, Aarya Parekh, declare that this is a self-initiated project undertaken to develop practical skills in machine learning, deep learning, and quantitative finance. All code and analysis is my own work.

Aarya Parekh

B.Tech Energy Engineering, Roll No. 24B1535

Date: _____